

System Context

Generalized Notation Notation (GNN) couples a small, declarative text language for generative models in the Active Inference lineage to a deterministic processing pipeline that turns each specification into validation, visualization, simulation, and analysis artifacts (Smékal and Friedman 2023). The language gives a model one canonical written form; the pipeline gives that form many executable and graphical realizations. This section describes both halves of the architecture and the way they meet, and it states the mathematical objects the language is obliged to carry — per model kind, because the language is not tied to a single shape of generative model.

The GNN Language

The GNN Language I

A GNN model is a plain-text document organized into named sections that together pin down the generative model its author intends. The `StateSpaceBlock` declares the variables of the model and their dimensions — hidden states, observations, control factors, policies, or the system matrices of a continuous model — establishing the shape of every tensor that follows. The `Connections` section records the directed and undirected dependencies among those variables, the edges of the underlying factor graph that downstream tools read to lay out diagrams and wire up inference. Which family of generative model a file declares is decided by the notation the file carries, not by its filename or its directory: the declared matrix vocabulary is what the pipeline reads to classify the model, as `sec. sec:model_kinds` specifies. The full construct vocabulary is catalogued in

The GNN Language I

tbl. 6.

Model Kinds

Model Kinds I

Every GNN file denotes one generative model, and the language lets that model take several distinct mathematical shapes. A file whose `StateSpaceBlock` declares the categorical tensors `A`, `B`, `C`, `D` — optionally `E` — denotes a discrete state-space generative model. A file that declares the system matrices `F`, `H`, `Q`, `R` together with a Gaussian prior over the initial state (`prior_mean`, `prior_cov`) denotes a continuous linear-Gaussian state-space model. Per-agent and per-level declarations (`A_agent1`, `A_level1`, ...) compose the categorical vocabulary into multi-agent and hierarchical structures. A file that declares boundary structure only — neither a categorical nor a continuous parameterization — is a structural wrapper: it names the model's interfaces but carries no renderable form.

Model Kinds I

The pipeline classifies each parsed specification structurally, and it does so from typed fields alone: the raw `## GNNSection` value, the declared matrix-key patterns, explicit agent counts, and explicit `## ModelParameters` keys. The classifier (`detect_model_kind` in `src/gnn/render/pomdp_contract.py`) resolves kinds in a fixed precedence order — multi-agent, then hierarchical, then continuous, then learning, then factored, then structural, then the flat single-factor case — so the more specific structure wins when a file exhibits several at once. Two properties of this classification matter for everything downstream. First, it is structural, not textual: prose in a model name or annotation never changes how a model renders, so a passing mention of “Dirichlet” in free text cannot reroute a specification into the parameter-learning kind. Second, it is computed

from the parsed specification, not asserted by the author: the same file classifies the same way on every machine, which is what lets rendering and execution behavior be stated per kind rather than per file. The taxonomy, its notation blocks, exemplar folder, and rendering and execution reach per kind are enumerated in tbl. 1.

Model Kinds I

Table 1: The model kinds the pipeline represents and executes, one row per kind. Notation blocks are the parameterization each kind declares; exemplar folders are the committed corpus under `input/gnn_files/`. `Renderer(s)/Executor(s)` cells are generated from the registry flags (see tbl. 4): the two parameterization rows carry their own flag sets, and multi-agent inherits the discrete row's coverage because its per-agent matrix keys canonicalize through the same discrete A/B/C/D render path; `recursive/` is reserved for bounded --autonomous proposal-loop runs and holds no committed models.

Model kind	Notation block	Exemplar folder	Renderer(s)	Executor(s)
Discrete categorical	A/B/C/D/[E] (column-stochastic B slices)	<code>input/gnn_files/discrete/</code>	all 9 registry frameworks	all 9 registry frameworks
Continuous linear-Gaussian	F/H/Q/R + priors (optional closed-loop pair)	<code>input/gnn_files/continuous/</code>	RxInfer.jl, JAX, PyTorch, NumPyro, Stan	RxInfer.jl, JAX, PyTorch, NumPyro, Stan
Multi-agent	<code>nr_agents + per-agent matrix keys (A_agent1, ...)</code>	<code>input/gnn_files/multiagent/</code>	all 9 registry frameworks	all 9 registry frameworks
Recursive	—	<code>input/gnn_files/recursive/</code>	—	—

The Discrete Categorical Kind I

A discrete specification denotes a partially observable Markov decision process over a horizon T , factorized as in eq. 1 following the standard formulation (Da Costa et al. 2020; Smith et al. 2022).

$$P(o_{1:T}, s_{1:T}, \pi) = P(\pi) P(s_1) \prod_{t=1}^T P(o_t | s_t) \prod_{t=2}^T P(s_t | s_{t-1}, \pi) \quad (1)$$

The four matrices named in the `StateSpaceBlock` are exactly the factors of eq. 1, and this correspondence is what makes the notation translatable rather than merely descriptive. The A matrix is the likelihood, a column-stochastic map from hidden states to observation outcomes, given in eq. 2.

The Discrete Categorical Kind I

$$P(o_t = i \mid s_t = j) = A_{ij}, \quad \sum_i A_{ij} = 1 \quad (2)$$

The B matrix is the controlled transition dynamics: one column-stochastic slice per action, as in eq. 3. The canonical reading of the tensor is column-stochastic, `B[next_state][previous_state][action]`, matching the convention of the discrete message-passing libraries the pipeline targets (Heins et al. 2022); validation records every declared tensor's detected orientation and can transpose row-stochastic textbook literals on request, as described in sec. 4.

The Discrete Categorical Kind I

$$P(s_{t+1} = i \mid s_t = j, u_t = k) = B_{ij}^{(k)}, \quad \sum_i B_{ij}^{(k)} = 1 \quad (3)$$

The C vector holds log-preferences over observations, which enter inference as the biased outcome distribution of eq. 4; the D vector is the prior over initial hidden states, eq. 5.

$$\tilde{P}(o_t = i) = \sigma(C)_i = \frac{\exp C_i}{\sum_j \exp C_j} \quad (4)$$

$$P(s_1 = i) = D_i, \quad \sum_i D_i = 1 \quad (5)$$

Given those factors, perception is approximate Bayesian inference: a variational posterior $Q(s)$ is fitted by minimizing the variational free energy of eq. 6, which decomposes into a complexity term and an accuracy term (Friston 2010).

The Discrete Categorical Kind I

$$F[Q] = \underbrace{D_{\text{KL}}[Q(s) \| P(s)]}_{\text{complexity}} - \underbrace{\mathbb{E}_{Q(s)}[\ln P(o | s)]}_{\text{accuracy}} \quad (6)$$

Action is expected-free-energy minimization: each policy π is scored over future steps τ by eq. 7, whose two terms are the information gain a policy is expected to yield and the extent to which its predicted outcomes match $\mathcal{C}$ (Da Costa et al. 2020).

The Discrete Categorical Kind I

$$G(\pi) = - \underbrace{\mathbb{E}_{Q(o_\tau, s_\tau | \pi)} [\ln Q(s_\tau | o_\tau, \pi) - \ln Q(s_\tau | \pi)]}_{\text{epistemic value}} - \underbrace{\mathbb{E}_{Q(o_\tau | \pi)} [\ln \tilde{P}(o_\tau)]}_{\text{pragmatic value}} \quad (7)$$

The policy posterior then combines those scores with the habit prior E declared in the specification, as in eq. 8, and the selected action u is sampled from it.

The Discrete Categorical Kind I

$$Q(\pi) = \sigma(\ln E - G) \tag{8}$$

The discrete categorical kind is the language's original and most densely exercised target: 27 of the 30 exemplar specifications in the corpus are discrete-state. The discrete family folder holds the kind's core set — from minimal `markov_chain.md` and `hmm_baseline.md` fixtures through epistemic planning (`tmaze_epistemic.md`) and deep planning horizons (`deep_planning_horizon.md`) to the canonical full agent (`actinf_pomdp_agent.md`) — and further categorical fixtures populate the basics, precision, structured, gridworld, and scaling-study corpora. The kind composes upward as well: multiple independent state factors (a factored model), per-level and per-agent matrix declarations (`sec. sec:structural_variants`), and Dirichlet priors that turn a declared matrix into a learned latent (`sec. 7`) are all extensions of this same

The Discrete Categorical Kind I

categorical substrate.

Recent work continues to refine how expected-free-energy objectives relate to variational inference and to alternative but equivalent formulations, which is why GNN keeps the objects of eq. 1 through eq. 8 explicit in the notation rather than burying them in backend-specific code (Champion et al. 2024; Nuijten, Lukashchuk, et al. 2026; Nuijten, Laar, et al. 2026). A `ModelParameters` section fixes the scalars those objects depend on — factor cardinalities and precision terms — while a `Time` section declares whether the model is static or dynamic and, if dynamic, how the horizon T and its discretization are organized. Because every one of these sections is explicit text, a GNN file is at once human-readable, diffable under version control, and unambiguous to a parser — the property that lets the rest of the

The Discrete Categorical Kind I

pipeline operate deterministically.

The Continuous Linear-Gaussian Kind I

A file whose `StateSpaceBlock` declares F , H , Q , R together with `prior_mean` and `prior_cov` denotes a continuous-state linear-Gaussian state-space model stepped in discrete time: a Gaussian latent state $x_t \in \mathbb{R}^n$ evolves linearly, and a Gaussian observation $y_t \in \mathbb{R}^m$ reads it out linearly.

$$x_1 \sim \mathcal{N}(\mu_0, \Sigma_0), \quad x_t = F x_{t-1} + u_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q) \quad (9)$$

The Continuous Linear-Gaussian Kind I

$$y_t = H x_t + v_t, \quad v_t \sim \mathcal{N}(0, R) \quad (10)$$

Here F is the state-transition matrix, H the observation matrix, Q the process-noise covariance, and R the observation-noise covariance; `prior_mean` (μ_0) and `prior_cov` (Σ_0) fix the Gaussian prior over the initial state, so the model's randomness is declared entirely in its parameterization rather than in categorical tables. The shape contract mirrors the semantics: `prior_mean` must carry exactly n entries, and Q , R , and `prior_cov` must be $n \times n$, $m \times m$, and $n \times n$ respectively — a mismatch is a validation-time diagnostic in the notation, not a runtime surprise inside a numerical backend.

The Continuous Linear-Gaussian Kind I

The kind also carries an optional closed-loop control structure. When a specification declares `goal_mean` ($\mu^\star$, the preferred state) and `control_gain` (k , a scalar proportional gain) alongside a control input u_t , the rendered model closes the loop on beliefs: the controller steers the filtered posterior mean toward the preferred state.

$$u_t = k(\mu^\star - \mu_t), \quad \mu_t = \mathbb{E}[x_t | y_{1:t}] \tag{11}$$

The Continuous Linear-Gaussian Kind I

This is how `input/gnn_files/continuous/continuous_navigation.md` is built: a two-dimensional navigator whose position persists between steps ($F = I$), whose observation is a noisy identity readout of that position, and whose control input pushes the believed position toward `goal_mean` with gain `control_gain`. The other continuous exemplars are passive: `predictive_coding_agent.md` and `stochastic_dynamics.md` filter without steering. In total the corpus ships 3 continuous linear-Gaussian exemplars in the continuous family folder. Symbolically the kind reuses letters the discrete vocabulary already claims — most prominently F , which names the state-transition matrix here and the variational free energy in eq. 6 — and sec. 9 resolves that collision by scoping each symbol table to its kind.

Multi-Agent, Hierarchical, and Structural Kinds I

The remaining kinds are structural compositions of the two parameterization vocabularies above, and the classifier recognizes each by its declared key pattern rather than by convention.

Multi-agent. A specification with an explicit agent count greater than one, or with per-agent matrix keys (A_agent1, B_agent1, C_agent1, ...), declares a multi-agent model: each agent carries its own categorical vocabulary, and coordination is expressed in the Connections section rather than by merging the agents into one joint table. `input/gnn_files/multiagent/stigmergic_swarm.md` is the worked exemplar — three agents coordinating through environmental traces (stigmergy) on a shared grid, with no direct communication between them — alongside `multi_agent_coordination.md` and a compact

clustered mean-field acceptance fixture.

Hierarchical. Per-level matrix keys (`A_level1`, `B_level1`, `A_level2`, ...) declare a multi-level model in which one level's beliefs supply another level's context — `input/gnn_files/hierarchical/hierarchical_pomdp.md` couples a fast lower level to a slow higher level that switches its context, and `temporal_hierarchy.md` develops the same structure over time. For categorical backends the per-level matrices are composed into one joint POMDP; the `RxInfer.jl` renderer additionally carries native strategies for two-level models, so a hierarchy can render either composed or natively depending on the target.

Learning and factored. A specification that declares a Dirichlet prior over one of its matrices (`dirichlet_A, ...`) — or sits in a declared learning corpus — is the parameter-learning kind: a matrix becomes a latent variable inferred from observations rather than a fixed table. `input/gnn_files/learning/dirichlet_likelihood_learning.md` is the worked exemplar: its `StateSpaceBlock` still declares `A[3,3]` in the categorical vocabulary, but `A` is a latent variable with prior pseudo-counts declared in `dirichlet_A[3,3]`, and hidden-state inference and likelihood learning run jointly by variational message passing with a mean-field factorization. The file is explicit about what its `InitialParameterization` means — the numeric `A` there is the environment's ground-truth likelihood that the agent never

Multi-Agent, Hierarchical, and Structural Kinds I

observes and must recover, not the agent's belief — and its annotation records why the Dirichlet prior is identity-biased rather than uniform: a fully uniform prior leaves the column-permutation symmetry unbroken and inference converges to a label-switched optimum. A specification whose `## ModelParameters` declare more than one independent state factor without per-level or per-agent keys is the factored kind, the standard multi-factor categorical POMDP.

Structural. A file that declares boundary structure only — a Markov-blanket-style wrapper with neither a categorical $A/B/C/D[E]$ parameterization nor a continuous $F/H/Q/R$ one — is classified structural and is render-only and informational: it names interfaces the surrounding system can reason about, and every framework reports it `unsupported` with the reason that it has no renderable form, rather than being forced through a discrete render path that would fail on missing matrices.

The recursive corpus. The `input/gnn_files/recursive/` directory is reserved for bounded `--autonomous` proposal-loop runs and holds no committed models; recursion today enters the taxonomy as structure — a level referencing its own kind, as in the hierarchical compositions above — rather than as a separately renderable kind. The corpus directory is part of the tree so that autonomous runs have a bounded, inspectable home, and its emptiness is a fact of the current release rather than an omission of the documentation.

Reading Exemplars by Kind I

Each kind is best understood through a specification that exercises it, and the corpus pairs every kind with at least one canonical exemplar. Reading one file per kind side by side makes the notation-block difference — the entire content of the taxonomy — concrete.

The discrete canonical is `input/gnn_files/discrete/actinf_pomdp_agent.md`, a fully controllable one-factor POMDP. Its `StateSpaceBlock` declares `A[3,3]`, `B[3,3,3]`, `C[3]`, `D[3]`, and `E[3]` — one observation modality, one hidden-state factor, and three actions — and every tensor of eq. 1 through eq. 8 is present by name: `A` carries the likelihood mapping, `B` carries the controlled transitions in the canonical column-stochastic slice order,

C holds log-preferences, D the initial-state prior, and E the habit prior. The `Connections` section then wires the inference cycle explicitly — $D \rightarrow s$ conditions the initial belief, $s \rightarrow A$ ties states to observations, $C \rightarrow G$ and $E \rightarrow \pi$ feed preference and habit into policy scoring, and $\pi \rightarrow u$ closes the loop from policy to action — so the file reads as a literal transcription of eq. 6 and eq. 7 into declared variables and edges.

The continuous canonical is `input/gnn_files/continuous/continuous_navigation.md` (eq. 9 through eq. 11). Its `StateSpaceBlock` declares no categorical tensors at all: $x[2,1]$ and $y[2,1]$ are the Gaussian state and observation, $F[2,2]$, $H[2,2]$, $Q[2,2]$, and $R[2,2]$ carry the linear dynamics and noise, `prior_mean[2]` and `prior_cov[2,2]` fix the initial belief, and `goal_mean[2]` and `control_gain[1]` add the closed-loop control structure. The `InitialParameterization` gives the matrices numerically — identity dynamics and identity readout, diagonal process and observation noise, a prior centered on the origin, a preferred position, and the proportional gain — and the `Equations` section restates

eq. 9, eq. 10, and eq. 11 as the model's declared dynamics. Nothing about the file is a discretized stand-in for a POMDP; it is a linear-Gaussian model by declaration, which is what lets the classifier label it CONTINUOUS and the continuous-capable backends render it natively.

The composed kinds show their structure in their key names. `input/gnn_files/multiagent/stigmergic_swarm.md` declares `A_agent1[4,9]`, `B_agent1[9,9,4]`, `C_agent1[4]`, and `D_agent1[9]`, then repeats the block for each further agent, so the per-agent pattern that the classifier keys on is visible directly in the `StateSpaceBlock`; coordination between the agents is expressed through shared environment variables in `Connections`, not by merging the agents into one joint table. `input/gnn_files/hierarchical/hierarchical_pomdp.md` declares

`A_level1[4,4]` and `A_level2[4,2]` with matching per-level transitions and priors — a fast lower level whose beliefs a slow higher level contextualizes — and it is these per-level keys that let the categorical backends compose the levels into one joint POMDP while the `RxInfer.jl` strategies render the two levels natively. In every case the classification is a mechanical reading of the declared names, which is the property that keeps the taxonomy honest: a file is its kind because of what it declares, not because of where it sits.

What the Kind Classification Governs I

The classification is not an annotation a report displays and forgets; it governs observable pipeline behavior at each stage that touches a model. At the render step, the per-framework receipts group outcomes into three disjoint sets — frameworks processed, frameworks failed, and frameworks unsupported — and the unsupported set is excluded from the success denominator rather than counted against it: a categorical backend facing a continuous model, or any backend facing a structural wrapper, reports a status with a reason and drops out of the rate, so a pipeline run over a mixed-kind corpus neither inflates its failures nor hides its gaps. At the execute step the same discipline applies, with skipped and degraded lanes recorded as explicit statuses. At the analysis and reporting steps, the receipts keep the distinctions

What the Kind Classification Governs I

legible: a continuous model whose categorical-only lanes report unsupported by capability reads differently from a discrete model whose continuous attempt failed, and the per-family acceptance ledger that the model-family gate writes carries those statuses per family, per backend, per step.

The governance runs in the direction the taxonomy promises: classification is computed once from the parsed specification and consumed everywhere downstream, so the render receipts, the execution traces, and the acceptance ledger all describe the same kind for the same file. A reader auditing any of those artifacts can recompute the classification from the file's declared vocabulary alone — tbl. 1 is the key — and expect to match what the pipeline recorded, which is what makes the kind taxonomy a checkable claim rather than a documentation convention.

The Processing Pipeline

The Processing Pipeline I

The pipeline is a fixed sequence of 25 numbered steps, 0–24, each a self-contained stage that consumes the artifacts of its predecessors and writes typed outputs for those that follow. Early steps parse and type-check the GNN text and validate it against the language schema; middle steps render visualizations, export the model to executable backends, and run simulations; later steps perform analysis, reporting, and downstream integration. The data dependencies among the steps form the directed acyclic graph shown in fig. 1, which makes the whole flow inspectable: any artifact can be traced back to the step that produced it and forward to every step that depends on it.

The Processing Pipeline I

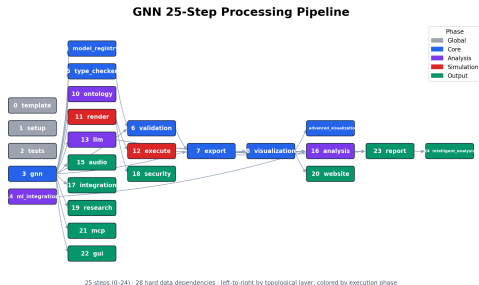


Figure 1: The 25-step GNN processing pipeline as a directed acyclic graph, laid out left to right by topological layer and colored by execution phase (legend, upper right). Each node is the thin step orchestrator under `src/gnn/` named in tbl. 2; each arrow is a hard data dependency parsed from the Data Dependency Graph block of `src/gnn/STEP_INDEX.md`, so any artifact can be traced back to the step that produced it. Parsing (Step 3) fans out to nearly every later step, and analysis (Step 16) collects enrichment from execution, visualization, and the LLM step. The figure is generated from `src/gnn/STEP_INDEX.md` at commit `dd6fb8ee3`; its footer restates the step count, the number of hard dependencies, and the layout rule.

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The per-step responsibilities are enumerated in tbl. 2; each row names a step and the transformation it owns within the 0–24 range.

The Processing Pipeline I

Table 2: The pipeline steps with the thin orchestrator module that owns each and the one-line purpose parsed from the master table in `src/gnn/STEP_INDEX.md`. Read the Step column against the arrows of fig. 1: each row's purpose is the transformation that module owns, and the table is regenerated from the index at every build, so it cannot drift from the code it describes.

Step	Module	Purpose
0	<code>0_template.py</code>	Pipeline template & initialization
1	<code>1_setup.py</code>	Environment setup & UV dependency install
2	<code>2_tests.py</code>	Test suite execution (pytest)
3	<code>3_gnn.py</code>	GNN file discovery & multi-format parsing
4	<code>4_model_registry.py</code>	Model versioning & registry management
5	<code>5_type_checker.py</code>	GNN type validation & resource estimation
6	<code>6_validation.py</code>	Consistency & semantic quality checking
7	<code>7_export.py</code>	Multi-format export (JSON, XML, GraphML, GEXF, Pickle)
8	<code>8_visualization.py</code>	Graph & matrix visualization generation
9	<code>9_advanced_viz.py</code>	Interactive / advanced visualization (Plotly, D3)
10	<code>10_ontology.py</code>	Active Inference ontology processing & validation
11	<code>11_render.py</code>	Code generation for simulation frameworks
12	<code>12_execute.py</code>	Execute rendered simulation scripts
13	<code>13_llm.py</code>	LLM-enhanced analysis & model interpretation
14	<code>14_ml_integration.py</code>	Machine learning integration & model training
15	<code>15_audio.py</code>	Audio sonification generation (SAPF)

The Processing Pipeline II

16	16_analysis.py	Statistical analysis &
17	17_integration.py	cross-simulation aggregation
18	18_security.py	System integration &
19	19_research.py	cross-module coordination
20	20_website.py	Security validation & generated
21	21_mcp.py	code scanning
22	22_gui.py	Research tools & literature
23	23_report.py	references
24	24_intelligent_analysis.py	Static HTML website generation
		Model Context Protocol
		processing & tool registration
		Interactive GNN constructor GUI
		Comprehensive analysis report
		generation
		AI-powered pipeline analysis &
		executive reports

The Processing Pipeline I

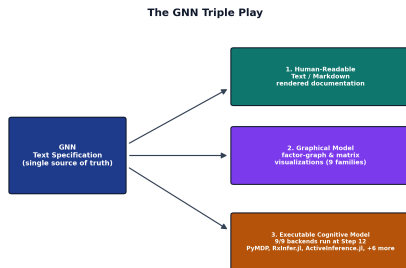
This staged design keeps the architecture modular. The implementation is organized into 44 source packages, one cluster of responsibilities per concern, and is documented across 624 documentation files so that each step's contract, inputs, and outputs are specified independently of the others. New backends or analyses attach to the graph by declaring their dependencies rather than by editing a monolith, and the deterministic step ordering means a model processed today yields the same artifacts when reprocessed tomorrow. The pipeline is kind-agnostic in its plumbing and kind-aware at its edges: parsing, validation, and export operate on whatever the file declares, while the rendering and execution steps consult the classification of `sec. sec:model_kinds` to decide which backends a given model can reach and record, explicitly, the ones it cannot.

The Triple Play

The Triple Play I

The reason for separating a single written language from a multi-stage pipeline is the design goal GNN calls the Triple Play: one model specification, three coordinated modes of existence. The same GNN text is simultaneously a human-readable model description, a set of graphical visualizations of its state space and factor structure, and an executable cognitive model that can be run as a simulation. fig. 2 depicts these three faces and the shared specification at their center.

The Triple Play I



One specification, three renderings — driven through the 25-step pipeline (steps 0-24).

Figure 2: The Triple Play: one GNN text specification (left) and its three coordinated projections (right) — the human-readable text form, graphical model visualizations, and executable simulation code. All three arrows originate at the same source box because all three projections are generated from one parsed document, which is why they cannot drift apart; the executable node states that 9 of 9 registered backends run at Step 12 and previews three of them by name. Counts are producer tokens at commit dd6fb8ee3, and the panel is generated by `scripts/manuscript_fig_triple_play.py`. Take-away: a model is authored once and then read, seen, and run from that single source.

The Triple Play I

Each face is generated from the same source, so they cannot drift apart. The text is the contract a researcher reads and reviews; the visualizations expose the factor-graph structure for inspection and communication; and the executable rendering lets the identical model be simulated across inference frameworks, connecting the written specification to the discrete message-passing libraries for categorical Active Inference (Heins et al. 2022), to the reactive message-passing and probabilistic-programming engines that carry the linear-Gaussian kind, and to the broader tooling ecosystem for compositional model representation (Felice et al. 2021). Because all three derive from one parsed document, GNN turns a model from a static artifact into a live object that can be read, seen, and run without re-encoding it for each purpose (Smith et al. 2022). The Triple Play holds per kind: each of the three faces is generated from the same classification the pipeline

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computed at parse time, so the diagram a reader sees, the notation a reviewer diffs, and the program a backend executes are all projections of one declared model kind rather than three loosely coupled guesses about what the file meant.